

12.HANGMAN PROJECT

Aim:

Write a c program to demonstrate hangman project using function, control statement, string, array, loops and timer mechanism.

Description:

The Hangman Game is a C-based interactive word-guessing program designed using fundamental programming concepts such as arrays, strings, loops, conditional statements, functions, and timer mechanisms. The game randomly selects a secret word from a predefined list, and each word is associated with a helpful hint to assist the player. All characters in the selected word are hidden using underscores, and the player must guess the word letter by letter before exceeding the allowed number of wrong attempts. The program also includes a graphical text-based hangman drawing that update with every incorrect guess, providing a visual representation of the game's progress. The game begins by asking the user to enter their name and then displays the hint related to the chosen word. A real-time countdown timer of 50 seconds is added to create urgency and improve user engagement. If the player does not complete the word within the time limit, the game automatically ends, showing that the hangman is fully drawn. Each guess is checked using loops, and the program enter the correct letter, it reveals correct letters in their respective positions. Incorrect guesses increase the wrong attempt counter, which updates the hangman drawing step-by-step using a separate function. The game continues until the word is correctly guessed. It ends when time ends or they enter out of chances. The project also demonstrates input validation, such as preventing repeated guesses of the same letter and handling both correct and incorrect inputs efficiently. The modular structure, including separate functions for drawing the hangman, checking letters, and managing the timer, makes the code easy to read and maintain. Overall, this Hangman game

project highlights how core C programming concepts can be combined to create an interactive, time-based game. It enhances understanding of loops, conditions, arrays, string handling, random word generation, and function-based program design, making it a valuable learning project for beginners in C.

List of Modules:

- Main module
- Game logic module
- Word management module
- Display module

Modules:

1.Main module

- It Control the overflow of the program
- Calls the appropriate functions such as starting the game, showing rules, or exiting
- Looping the menu until the user quits
- Calling game module to start Hangman

2.Game logic module

- It handles the entire game logic of the program
- Selecting the secret word
- Taking letter input from the user
- Checking whether the letter is correct or wrong
- Updating the hidden word with correct guesses
- Maintaining the number of wrong attempts
- Determining win or loss
- Calling the drawing module

3.Word management module

- It manages the list of words in a game.
- Storing words in an array or file
- Selecting a random word for the game
- Sending the selected word to the game module

4.Display module:

- It draws the hangman figure stage by stage
- Display the hangman at each wrong attempt
- It displays correct guessed letter and remaining attempts
- It prints final win or lose statement

Main function:

- initializes game
- selects random word
- takes user input
- controls the timer
- checks win/lose status
- draw (int wrong)

Library function:

- printf() -- print output
- scanf() -- read user input
- strlen() -- length of string
- strcpy() -- copy selected word
- strcmp() -- compare guessed word
- srand() -- random number seed
- rand() -- generate random index
- time() -- get current time
- difftime() -- calculate time difference

Program:

```
#include <stdio.h>

#include <string.h>

#include <stdlib.h>

#include <time.h>

#define MAX 50

void draw(int wrong) {

printf("\n");

printf(" +---+\n");

printf(" | \n");
```

```

if (wrong >= 1)
printf(" O \n");
else
printf(" \n");
if (wrong == 2)
printf(" | \n");
else if (wrong == 3)
printf(" /| \n");
else if (wrong >= 4)
printf(" /\ \n");
else
printf(" \n");
if (wrong == 5)
printf(" / \n");
else if (wrong >= 6)
printf(" / \ \n");
else
printf(" \n");
printf(" \n");
printf("=====\n");
}

int main() {

char words[15][MAX] = {"guava", "hippopotamus", "joystick", "pizza",
"beach","dictionary","calculator","diwali","pigeon","white","kitkat","biryani","
anirudh","dhoni","viratkohli"};

char hints[15][MAX] = {"A fruit", "Wild animal", "Electronic device", "Italian
food", "Place near sea", "word book", "device to perform arithmetic
operations", "festival of lights", "bird of peace", "symbol of purity ", "chocolate-

```

```

TAKE A BREAK", "king of all foods", "The Rockstar of music", "captain
cool", "jersey no.18"};
char guessed [20];
int wrong = 0, maxWrong = 6;
char guess;
int i, found;
srand(time(NULL));
int index = rand() % 15;
char word[20];
strcpy(word, words[index]);
int wordLength = strlen(word);
for (i = 0; i < wordLength; i++)
    guessed[i] = '_';
guessed[wordLength] = '\0';
printf("\n Welcome to Hangman!\n");
char name[20];
printf("Enter the player name:");
scanf("%s", name);
printf(" Hint: %s\n", hints[index]);
time_t start, now;
int timeLimit = 50; // 50 seconds timer
time(&start);
while (wrong < maxWrong) {
    time(&now);
    int timeLeft = timeLimit - (int)difftime(now, start);
    if (timeLeft <= 0) {
        printf("\n Time's up!\n");
        printf(" Game Over! The man is HANGED!\n");
    }
}

```

```
printf("The word was: %s\n", word);
return 0;
}
printf("\n Time left: %d seconds\n", timeLeft);
draw(wrong);
printf("\nWord: %s\n", guessed);
printf("Enter a letter: ");
scanf(" %c", &guess);
found = 0;
for (i = 0; i < wordLength; i++) {
if (word[i] == guess && guessed[i] == '_') {
guessed[i] = guess;
found = 1;
}
}
if (!found) {
wrong++;
printf(" Wrong! Chances left: %d\n", maxWrong - wrong);
} else {
printf(" Correct!\n");
}
if (strcmp(word, guessed) == 0) {
draw(wrong);
printf("\n You WIN! The word is: %s\n", word);
return 0;
}
}
```

```
draw(wrong);

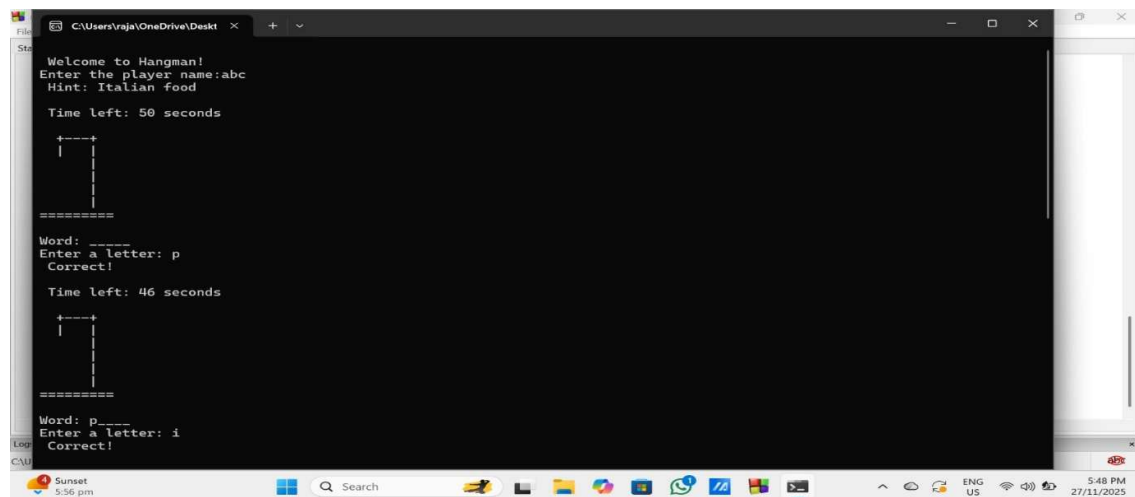
printf("\n Game Over!\nSorry %s!!! You are HANGED!\n", name);

printf("The word was: %s\n", word);

return 0;

}
```

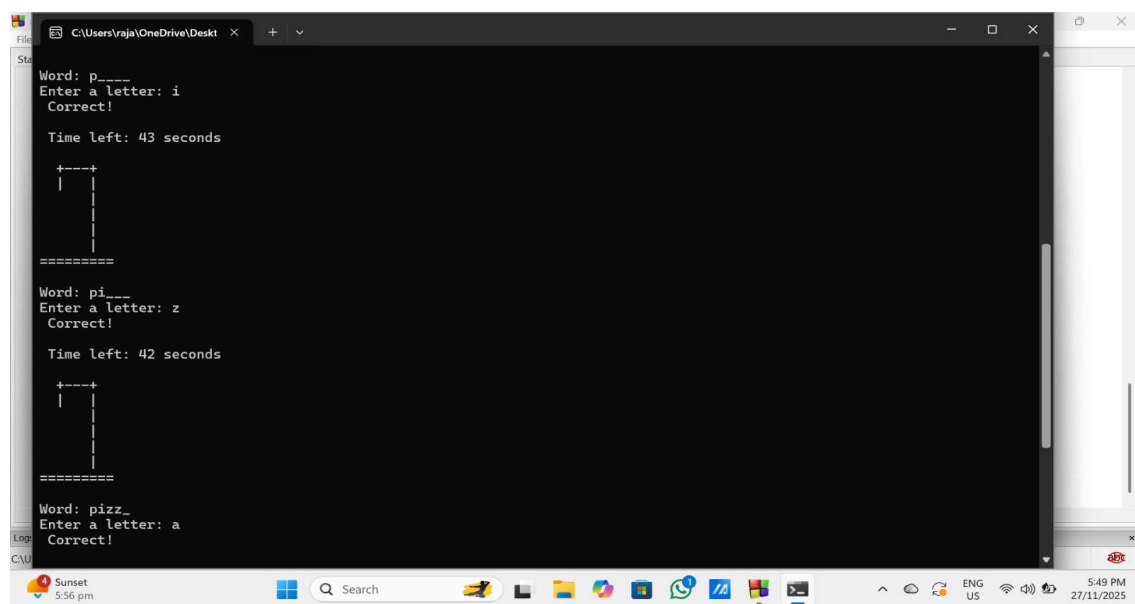
OUTPUT:



```
Welcome to Hangman!
Enter the player name: abc
Hint: Italian food
Time left: 50 seconds

+---+
|
|
|
=====
Word: _____
Enter a letter: p
Correct!
Time left: 46 seconds

+---+
|
|
|
=====
Word: p____
Enter a letter: i
Correct!
```



```
Word: p____
Enter a letter: i
Correct!
Time left: 43 seconds

+---+
|
|
|
=====
Word: pi____
Enter a letter: z
Correct!
Time left: 42 seconds

+---+
|
|
|
=====
Word: pizz_
Enter a letter: a
Correct!
```

```
File Edit View Help
C:\Users\raja\OneDrive\Deskt x + -
Sta
Enter a letter: z
Correct!

Time left: 42 seconds

+---+
|   |
+---+

=====

Word: pizz_
Enter a letter: a
Correct!

+---+
|   |
+---+

=====

You WIN! The word is: pizza

Process returned 0 (0x0)   execution time : 14.193 s
Press any key to continue.

Log
CAU
```

Sunset 5:56 pm

Search

ENG US

5:49 PM 27/11/2025

```
miniproject.c - Code:Blocks 25.03
File Edit View Help
C:\Users\raja\OneDrive\Deskt x + -
Sta
Welcome to Hangman!
Enter the player name:xyz
Hint: word book

Time left: 50 seconds

+---+
|   |
+---+

=====

Word: _ _ _ _ _
Enter a letter: e
Wrong! Chances left: 5

Time left: 45 seconds

+---+
| 0 |
+---+

=====

Word: _ _ _ _ _

Log
C:\Users\raja\OneDrive\Desktop\miniproject.c
```

C/C++

Windows (CR+LF) WINDOWS-1252 Line 50, Col 28, Pos 1257

Insert

Read/Write default

26°C Partly sunny

Search

ENG US

5:56 PM 27/11/2025


```
miniproject.c - Code:Blocks 25.03
C:\Users\raja\OneDrive\Desktop\miniproject.c
Correct!
Time left: 43 seconds
+---+
|   |
0   |
|   |
+---+

=====
Word: _____y
Enter a letter: l
Wrong! Chances left: 4
Time left: 41 seconds
+---+
|   |
0   |
|   |
+---+

=====
Word: _____y
Enter a letter: p
Wrong! Chances left: 3
```

```
miniproject.c - Code:Blocks 25.03
C:\Users\raja\OneDrive\Desktop\miniproject.c
Enter a letter: p
Wrong! Chances left: 3
Time left: 38 seconds
+---+
|   |
0   |
/|   |
+---+

=====
Word: _____y
Enter a letter: d
Correct!
Time left: 35 seconds
+---+
|   |
0   |
/|   |
+---+

=====
Word: d_____y
Enter a letter: w
Wrong! Chances left: 2
```

```
miniproject.c - Code::Blocks 25.03
C:\Users\raja\OneDrive\Desktop
Enter a letter: w
Wrong! Chances left: 2

Time left: 33 seconds

+---+
|   |
| 0 |
|/ \|
|   |
+---+

=====

Word: d_____y
Enter a letter: j
Wrong! Chances left: 1

Time left: 30 seconds

+---+
|   |
| 0 |
|/ \|
|   |
+---+

=====

Word: d_____y
Enter a letter: b
Wrong! Chances left: 0
```

```
Wrong! Chances left: 0

+---+
|   |
| 0 |
|/ \|
|   |
+---+

=====

Game Over!
Sorry xyz!!! You are HANGED!
The word was: dictionary

Process returned 0 (0x0)   execution time : 25.240 s
Press any key to continue.
```